Conceptual combination, proved

Selective modification, graded membership, and over-extension

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The puzzle: over-extension

- People judge some things members of a combination but not of one of its parts: a *guppy* is a better *pet fish* than it is a *fish* or a *pet* (Hampton 1988).
- Fuzzy logic forbids this: membership in *A and B* is at most the minimum of the two.
- So combination is not intersection. What is it?

One answer (Smith, Osherson, Rips & Keane 1988): the adjective *modifies* the noun's prototype. Typicality and membership follow from similarity to the modified prototype.

Prototypes (SO1, SO2)

apple		
diag.	attribute	votes
1	color	red 25, green 5
.5	shape	round 15, other 5
.25	texture	smooth 25, rough 5

- A **prototype**: attributes, each with a *diagnosticity*, and *votes* (salience) on its values.
- An **instance**: votes too, no diagnosticity. A red apple has all 30 color votes on red.

Typicality is similarity (SO3)

Tversky's contrast rule, attribute by attribute, each vote a feature:

$$\mathrm{Sim}(P, I) = \sum_i d_i \left[a\,|P \cap I|_i - b\,|P - I|_i - c\,|I - P|_i \right]$$

- Red apple in *apple*, on color: 25 shared, 5 green only in the prototype, 5 red only in the instance: $1 \cdot (25 - 5 - 5) = 15$.
- With $a = b = c = 1$, the red apple's typicality is $15 + 2.5 + 3.75 = 85/4$. The paper prints 21.

Selective modification (SO4)

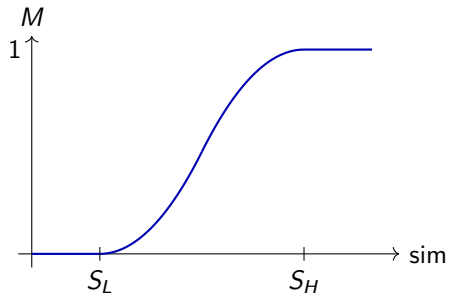
An adjective naming one value of one attribute (*red*: color):

1. **selects** that attribute of the noun;
2. **shifts** all its votes to the value: *red apple* has color red 30;
3. **boosts** its diagnosticity: by 2 in the paper ("an arbitrary choice"), a parameter here.

Every other attribute is unchanged.

In Lean: `modify S boost i0 v0 P`, in CogLib, with `sim_modify`: modification changes similarity only on the selected attribute.

Membership: the threshold model (H1)



- Hampton (2007): typicality and membership are both nondecreasing functions of *one* similarity.
- Membership is 0 below S_L, 1 above S_H, and two quadratic pieces between (his Eq. 8).
- Exactly rational: Lean computes it.

What is proved, for every parameter setting

For every prototype, every $a > 0$, $b, c \geq 0$, boost ≥ 1, a diagnostic attribute, and an instance with the adjective's value:

- **Conjunction effect**: the instance is more typical of *red apple* than of *apple*. It needs only that something change: a boost above 1, or noun votes not already all on red.
- **Reverse effect**: an instance with another value is *less* typical of *red apple*. It needs only $b + c > 0$.
- **The attribute, not the value**: a red apple in *red apple* is exactly as typical as a brown apple in *brown apple*, for every setting (p. 495).

From the conjunction effect to over-extension

Stated in CogLib for *any* similarity, not just this model:

- **Yes/no membership**: if the instance is more typical of the combination, then at some criterion it is judged a member of the combination and not of the noun.
- **Graded membership**: under any nondecreasing membership function, it is never less a member of the combination. Where the function rises, strictly more: over-extension.

So selective modification plus threshold membership *predicts* over-extension. Fuzzy logic's min rule is violated by construction.

Is it faithful? The paper's figures, exactly

Similarity (Figs. 1–4)	paper	Lean, exact
red apple in *apple*	21	85/4
brown apple in *apple*	−54	−215/4
bumpy red apple in *apple*	2.5	5/2
red apple in *red apple*	66	265/4
brown apple in *red apple*	−114	−455/4

The conjunction effect is 120 for *brown apple* and 45 for *red apple*, as on p. 495. Every value is proved by `decide +kernel`.

Numbers: concepts with a truth

- Attributes computed from the number: parity, last digit, digit sum mod 3, number of digits.
- The truth, proved: divisible by 6 $\iff$ even and divisible by 3; the digit-sum test for 3.
- **The prototype comes from experience**: *multiple of 3* counts the values of the exemplars $3, 6, \ldots, 30$.

multiple of 3	*even multiple of 3* (by SO4)	
parity: even 5, odd 5	parity: even 10	(diagnosticity 2)
last digit: 1 each	unchanged	
digit sum mod 3: 0×10	unchanged	
digits: one 3, two 7	unchanged	

14 is an even multiple of 3

	multiple of 3	*even multiple of 3*
typicality of 14	-41	-16
membership of 14	$81/1250 \approx .06$	$497/625 \approx .80$
membership of 9	$721/1250 \approx .58$	

- 14 is far more an *even multiple of 3* than a *multiple of 3*: **incoherent**.
- And 14 is not a multiple of 3: **false**. Lean proves both.
- In general: *every* even number is more typical of *even multiple of 3*, multiple of 3 or not.

$a = b = c = 1$, boost 2, Hampton bounds -50 and 0: chosen for illustration.

What this adds

- **Predictions for every setting**, with their conditions explicit, and the paper's numbers as a check.
- **Phenomena stated theory-neutrally** (CogLib): conjunction effect, over-extension. A rival theory is held to the same statements, as memory families are in the theory matrix.
- **A domain with truth**: over-extension separates into incoherence (judgments disagree with each other) and error (judgments disagree with the facts).

Limits, and next

- Hampton's membership function is computed but not yet proved monotone; the graded theorem assumes it rises.
- Diagnosticity is set to 1 for numbers; S&O derive it from a contrast class.
- Only single-attribute adjectives, as in the paper's precise model.
- **Rival theories**: view application (Shrager 1987), a very different account, combining abstract knowledge structures; relation linking for noun–noun combinations.
- Hampton's over-extension data, judged as evidence.